Thermodynamics (Brief Summary)

Describes thermal properties of macroscopic systems.

We begin by considering an isolated system — ideal gas — of N neutral atoms. Typically $N \gtrsim N_A = 6.022 \cdot 10^{23}$ (Avogadro constant) [eg. 12 grams of carbon contains one mole of C atoms, N_A.]

Equilibrium state of such a system is described by just a few <u>state variables</u>: volume V, pressure P, temperature T.

Quite generally (and deduced experimentally) there is an <u>equation of state</u>: $f(P,T,V)=0$ according to which the condition of a system is determined by fixing the values of <u>two</u> of the state variables.

<u>Example</u>: ideal gas

$$PV = \nu RT \quad (R = 8.314 \ J\,mol^{-1}K^{-1} \text{ is the gas constant})$$

ν = quantity of gas in moles

the same: $PV = N k_B T \qquad k_B = 1.38 \cdot 10^{-23} \ J/K$ = Boltzmann constant

Thus, P and V uniquely specify T, for example.

Then $f = \dfrac{PV}{\nu RT} - 1$.

<u>1st Law of Thermodynamics</u> (is just a conservation of energy)

In a closed system the total energy: mechanical, electromagnetic and thermal (heat) is constant.

* NB: one form of energy may be transformed into another, but the sum remains constant

(the Principle of the Conservation of Energy, R. Mayer 1842)

<u>Internal energy</u> $U = U(T,V)$ or $U(T,p)$ or $U(P,V)$ must be a single-valued function of the independent state variables, say T and V.

*NB U can not depend on how the state, characterized by T and V, is reached. It only depends on T and V values. Otherwise one would be able to gain energy by bringing the system from some initial (T_i, V_i) state to the final (T_f, V_f) via different paths on (T,V) plane.

This implies that the <u>differential</u>

$$dU = \left(\frac{\partial U}{\partial T}\right)_V dT + \left(\frac{\partial U}{\partial V}\right)_T dV$$

is the <u>exact differential</u> (total differential).

Contrary to that, the heat supplied to a system ($đQ$) is not <u>exact</u> differential: it depends on the <u>process</u> choosen. Similarly, the work done on the system ($đW$) is not exact differential.

However, the total change in internal energy

$$dU = đQ + đW \qquad \text{(First Law)}$$

For a <u>quasi-static</u> process $đW = -p\,dV$ (work done <u>on</u> the system)
(<u>minus</u> sign because compression ($dV<0$) increases U)

$$dU = đQ - p\,dV$$

or $$đQ = dU + p\,dV$$

Heat capacity

$$C = \lim_{dT \to 0} \frac{đQ}{dT}$$

where $đQ$ is the amount of heat needed to raise the temperature of the system by dT.

Thus $$C = \lim_{dT \to 0} \frac{dU + PdV}{dT}$$

depends on the process $đQ$ is supplied.

① <u>$V = \text{constant}$</u> $\Rightarrow dV = 0 \Rightarrow C_V = \lim_{dT \to 0} \frac{dU}{dT} = \left(\frac{\partial U(T,V)}{\partial T}\right)_V$

(partial derivative at constant volume)

② <u>$P = \text{constant}$</u> $\Rightarrow$ thermal expansion extracts work from the system $\Rightarrow$ <u>more heat</u> is needed to achieve dT.

$$C_P = \lim_{dT \to 0} \left(\frac{đQ}{dT}\right)_{P = \text{const}}$$

$$đQ = dU + PdV = \left(\frac{\partial U}{\partial T}\right)_V dT + \left(\frac{\partial U}{\partial V}\right)_T dV + PdV$$

$$= C_V\, dT + \left[\left(\frac{\partial U}{\partial V}\right)_T + P\right] dV.$$

Switch to (P, T) variables: $V = V(T, P)$

$$\Rightarrow dV = \left(\frac{\partial V}{\partial T}\right)_P dT + \left(\frac{\partial V}{\partial P}\right)_T dP$$

$\Rightarrow$ at constant pressure $\frac{dV}{dT} = \left(\frac{\partial V}{\partial T}\right)_P$

$$\Rightarrow C_P = C_V + \left[\left(\frac{\partial U}{\partial V}\right)_T + P\right]\left(\frac{\partial V}{\partial T}\right)_P$$

<u>Ideal gas</u> $\left(\frac{\partial U}{\partial V}\right)_T = 0$ because molecules <u>do not</u> interact with each other

$\Rightarrow$ energy can not depend on a distance between them.

Note $P\left(\frac{\partial V}{\partial T}\right)_P = \left(\frac{\partial (PV)}{\partial T}\right)_P = \frac{\partial}{\partial T}\left(\nu R T\right)_P = \nu R$

$\Rightarrow \quad C_p = C_V + \nu R$

$\Rightarrow$ <u>Heat capacity per mole = specific heat</u> (molar heat)

$c_p = c_v + R \qquad \left(c_p = \frac{C_p}{\nu}, \; c_v = \frac{C_v}{\nu}\right)$

(small letters)

$\boxed{c_p - c_v = R}$ for ideal gases : hold very well for He, Ar, Hg, O_2, CO...

*NB In solids and liquids $c_p \approx c_v$ because they hardly change their volumes on heating.

<u>Example : adiabatic process in ideal gas</u>

Adiabatic change $\Rightarrow \; đQ = 0$

Then $dU = đW = -P\,dV$ (note that this implies that now $đW$ is dW, an exact differential!)

$\Rightarrow \; dU + P\,dV = 0$

But $dU = c_v\,dT$, and $RT = PV \Rightarrow dT = \frac{1}{R}\,d(PV) = \frac{P\,dV + dP\,V}{R}$

$$\Rightarrow \quad \frac{c_v}{R}\left(P\,dV + V\,dP\right) + P\,dV = 0$$

$$\left(\frac{c_v}{R} + 1\right) P\,dV + \frac{c_v}{R}\,V\,dP = 0$$

$$\Rightarrow \quad c_p\,P\,dV + c_v\,V\,dP = 0$$

Divide by PV: $$c_p \frac{dV}{V} + c_v \frac{dP}{P} = 0$$

$$\frac{c_p}{c_v}\frac{dV}{V} + \frac{dP}{P} = 0$$

Denote $\gamma = \frac{c_p}{c_v} > 1$: $$\gamma\, d(\ln V) + d(\ln P) = 0$$

$$d\left(\ln V^{\gamma} \cdot P\right) = 0$$

$$\Rightarrow \boxed{V^{\gamma} P = \text{const}} \qquad \gamma = \frac{c_p}{c_v}$$

Ideal gas: $U = \frac{3}{2} RT$ (per mole) $\Rightarrow c_v = \left(\frac{\partial U}{\partial T}\right)_V = \frac{3}{2} R$

$$c_p = \left(\frac{3}{2}+1\right) R = \frac{5}{2} R.$$

$$\gamma = \frac{5}{2} / \frac{3}{2} = \frac{5}{3}$$

adiabat $\boxed{P V^{5/3} = \text{const}}$

Note that it is not <u>P = const</u> or <u>V = const</u> but their <u>function</u>.

<u>Question: what is heat capacity for an adiabatic process?</u>

Since $PV = RT \Rightarrow V = RT/P \Rightarrow PV^{5/3} = P\left(\frac{RT}{P}\right)^{5/3} = T^{5/3} P^{-2/3}$

$\Rightarrow T^{5/3} P^{-2/3} = \text{const}$ is also another way to represent it.

<u>Entropy: a first look</u>

$$đQ = dU + PdV$$

<u>inexact</u> differential.

However, can be used to construct an <u>exact one</u>!

Ideal gas: $U = \frac{3}{2} \nu RT \Rightarrow dU = \frac{3}{2} \nu R\, dT$

monoatomic

$$P = \nu RT/V \Rightarrow PdV = \nu RT \frac{dV}{V}$$

$$\Rightarrow \quad đQ = \frac{3}{2}\nu R\, dT + \nu R T \frac{dV}{V}$$

$$\frac{đQ}{T} = \frac{3}{2}\nu R \frac{dT}{T} + \nu R \frac{dV}{V}$$

$$\Rightarrow \quad \frac{đQ}{T} = \frac{3}{2}\nu R\, d(\ln T) + \nu R\, d(\ln V)$$

Thus $\frac{đQ}{T}$ is an <u>exact differential</u>

$$\int_i^f \frac{đQ}{T} = \frac{3}{2}\nu R \int_i^f d(\ln T) + \nu R \int_i^f d(\ln V)$$

$$= \frac{3}{2}\nu R \ln\frac{T_f}{T_i} + \nu R \ln\frac{V_f}{V_i} = \nu R \ln\left(\frac{T_f^{3/2} V_f}{T_i^{3/2} V_i}\right).$$

Determined by initial (T_i, V_i) and final (T_f, V_f) only: does not depend on a process that takes us from (i) to (f).

$\Rightarrow$ Define $S_f - S_i = \int_i^f \frac{đQ}{T}$ as the change in <u>entropy</u>

$$S_f = \nu R \ln\left(\frac{T_f^{3/2} V_f}{T_i^{3/2} V_i}\right) + S_i$$

Entropy S $\left(dS = \frac{đQ}{T}\right)$ is a new function of state for an ideal gas.

Can define it as $S = \int_i^f \frac{đQ}{T}$ with respect to some initial state (i).

Thus, $dS = \frac{đQ}{T} \Rightarrow dU = đQ - p\,dV$

$$\boxed{dU = T\,dS - p\,dV}$$ Thermodynamic identity.

Helmholtz free energy $F = U - TS$

Gibbs free energy $G = H - TS = U + PV - TS$

Enthalpy $H = U + PV$

* Second Law of Thermodynamics: the law of increase of entropy

The entropy of a closed system tends to remain constant or to increase with time when a constraint internal to the system is removed.

⇒ Heat engines, Carnot cycle, absolute temperature, and the rest of thermodynamics follow from entropy S.

But what is the physical meaning of it?

How to approach it / derive it / use it for non-ideal gases, solids, liquids, magnets and so on?

This is where <u>statistical Mechanics</u> begins.

A NOTE on Exact and Inexact differentials:

Suppose $G = G(x,y)$ is a continuous function.

Then $dG = \left(\frac{\partial G(x,y)}{\partial x}\right)_y dx + \left(\frac{\partial G(x,y)}{\partial y}\right)_x dy$

Re-write: $dG = \underset{\left(\frac{\partial G}{\partial x}\right)_y}{\underset{\parallel}{A(x,y)}}\, dx + \underset{\left(\frac{\partial G}{\partial y}\right)_x}{\underset{\parallel}{B(x,y)}}\, dy$

The order of mixed derivatives is not important

$$\Rightarrow \left(\frac{\partial}{\partial x}\left(\frac{\partial G(x,y)}{\partial y}\right)_x\right)_y = \frac{\partial^2 G(x,y)}{\partial x\,\partial y} = \frac{\partial^2 G(x,y)}{\partial y\,\partial x} = \left(\frac{\partial}{\partial y}\left(\frac{\partial G(x,y)}{\partial x}\right)_y\right)_x$$

That is, $\left(\frac{\partial B(x,y)}{\partial x}\right)_y = \left(\frac{\partial A(x,y)}{\partial y}\right)_x$.

When this is satisfied, dG is an <u>exact</u> differential.

<u>Example</u>: let $df = y^3 dx + (2y + y^2 x)\, dy$

$\Rightarrow A = y^3, \quad B = 2y + xy^2$

$\frac{\partial A}{\partial y} = 3y^2, \quad \frac{\partial B}{\partial x} = y^2 \Rightarrow df$ is ~~not~~ <u>inexact</u> differential.

Consider next $d\varphi = e^{xy}\, df$

$$d\varphi = e^{xy} y^3 dx + (2y + xy^2)\, e^{xy}\, dy$$

$$\Rightarrow \tilde{A} = y^3 e^{xy}, \quad \tilde{B} = (2 + xy)\, y\, e^{xy}$$

$$\frac{\partial \tilde{A}}{\partial y} = 3y^2 e^{xy} + y^3 x e^{xy} = (3 + yx)\, y^2 e^{xy}$$

$$\frac{\partial \tilde{B}}{\partial x} = \frac{\partial}{\partial x}(2y + xy^2) e^{xy} = y^2 e^{xy} + (2y + xy^2)\, y\, e^{xy}$$
$$= (3y^2 + xy^3)\, e^{xy}$$

But $\frac{\partial \tilde{A}}{\partial y} = \frac{\partial \tilde{B}}{\partial x}$ $\Rightarrow$ $đ\varphi = d\varphi$ is <u>exact</u> differential.

$\Rightarrow$ an <u>integrating factor</u> e^{xy} has turned inexact differential $đf$ into exact one, $d\varphi$.

<u>Back to physics</u>: $đQ$ is inexact differential,
but $dS = \frac{đQ}{T}$ is exact differential.

$\Rightarrow$ temperature T is an integrating factor.

useful to know that for a function of two variables (x, y) one can always find an integrating factor.